MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 3**

Continuity (Section 1.8) and limits at infinity (Section 3.4): the definition of continuity, the three ways it fails, building continuous functions, the Intermediate Value Theorem, horizontal asymptotes and the behaviour of a function far out.

Key facts to have at hand

- **Continuity at a number a:** $\lim_{x\to a} f(x) = f(a)$. Three things are hidden in this one equation: $f(a)$ is defined; the limit exists; the two are *equal*. If f is defined near a but not continuous at a, it is **discontinuous** at a.
- **Three ways to fail:** a *removable* discontinuity (the limit exists but the value is missing or wrong — one value repairs it); a *jump* (the one-sided limits exist and differ — no value repairs it); an *infinite* discontinuity (a one-sided limit is $\pm\infty$ — no value repairs it).
- **One-sided continuity:** continuous from the right at a means $\lim_{x\to a^+} f(x) = f(a)$; from the left, $\lim_{x\to a^-} f(x) = f(a)$. **Continuous on an interval** means continuous at every number inside it, and one-sidedly at any endpoint that belongs to it.
- **Building continuous functions:** if f, g are continuous at a then so are $f \pm g$, cf, fg and f/g *provided* $g(a) \neq 0$; if g is continuous at a and f at $g(a)$, then $f \circ g$ is continuous at a. A limit passes inside a continuous function: $\lim_{x\to a} f(g(x)) = f\big(\lim_{x\to a} g(x)\big)$.
- **Which functions:** polynomials, rational, root and trigonometric functions are continuous **on their domains** — and there "limit" means "substitute".
- **Intermediate Value Theorem:** if f is continuous on the *closed* interval $[a, b]$ and N is any number between $f(a)$ and $f(b)$ with $f(a) \neq f(b)$, then there is a number c in (a, b) with $f(c) = N$. It promises *at least one* such c and says nothing about where it is. Without continuity it is false.
- **Limits at infinity:** $\lim_{x\to\infty} f(x) = L$ means $f(x)$ is as close to L as we like for all x large enough. Then $y = L$ is a **horizontal asymptote** — which the graph is allowed to cross, even infinitely often.
- **The basic limit:** $\lim_{x\to\pm\infty} \frac{1}{x} = 0$, and $\lim_{x\to\infty} \frac{1}{x^r} = 0$ for every $r > 0$. All the limit laws hold for $x \to \pm\infty$ as they do for $x \to a$.
- **Method for a quotient:** divide the numerator and the denominator by the *highest power of x in the denominator*, then use $1/x^r \to 0$. For roots remember $\sqrt{x^2} = |x|$, so $\sqrt{x^2} = -x$ when $x < 0$: the sign is your job, not the algebra's.
- $\infty - \infty$ **and** ∞/∞ **are instructions to rewrite**, like $0/0$: divide, or multiply by the conjugate. $\lim_{x\to\infty} f(x) = \infty$ means $f(x)$ exceeds every number for x large enough.
- **Precise definition at infinity:** $\lim_{x\to\infty} f(x) = L$ means: for every $\varepsilon > 0$ there is a number N such that if $x > N$ then $|f(x) - L| < \varepsilon$. Same grammar as ε–δ: ε is the challenge, N the response.

Part I · Continuity and the Intermediate Value Theorem (Section 1.8) · 50 minutes

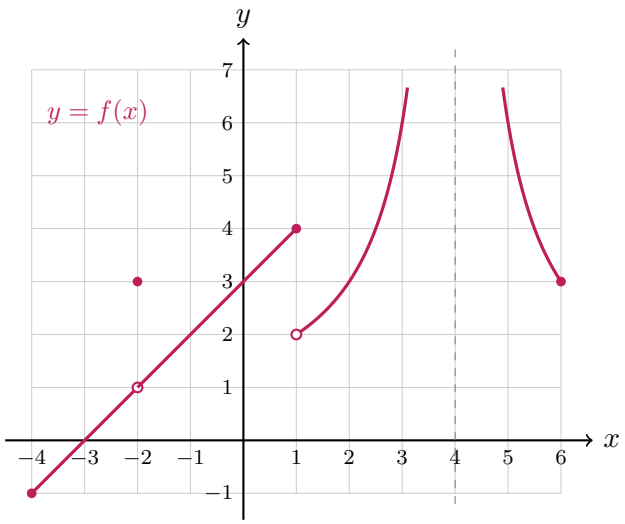
Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is continuous at a, then $\lim_{x\to a} f(x) = f(a)$.
2. If $\lim_{x\to a} f(x)$ exists, then f is continuous at a.
3. If $f(a)$ is defined and $\lim_{x\to a} f(x)$ exists, then f is continuous at a.
4. $f(x) = 1/x$ is a continuous function.
5. $f(x) = 1/x$ is continuous at 0.
6. If f and g are continuous at a, then f/g is continuous at a.
7. If $f(1) = -2$ and $f(3) = 5$, then $f(c) = 0$ for some c in $(1, 3)$.

8. If f is continuous on $[1,3]$, $f(1) = -2$ and $f(3) = 5$, then f has *exactly one* zero in $(1,3)$.
9. If f has a jump discontinuity at a, one can change the value $f(a)$ to make f continuous at a.
10. $f(x) = |x|$ is continuous at 0.

Problem 1 *Reading continuity off a graph.*

The graph of a function f is shown; the dashed line is a vertical asymptote. (f is defined at every number of $[-4,6]$ except $x = 4$.)



Find each of the following, or say that it does not exist.

(a) $\lim_{x\to -2^-} f(x)$ and $\lim_{x\to -2} f(x)$

(b) $f(-2)$

(c) $\lim_{x\to 1^-} f(x)$ and $\lim_{x\to 1^+} f(x)$

(d) $\lim_{x\to 4} f(x)$

(e) At which numbers is f discontinuous? Name the type of each discontinuity.

(f) At which of them can a change of one value make f continuous, and what value? On which intervals is f continuous?

Problem 2 *Where is each function continuous? Give the largest set, and name the theorem you use.*

(a) $f(x) = \dfrac{x^2 - 9}{x^2 - 4x + 3}$

(b) $g(x) = \sqrt{5 - x} + \sqrt{x + 1}$

(c) $h(x) = \dfrac{x + 2}{\sqrt{x^2 - 4}}$

(d) $k(x) = \sin\left(\dfrac{1}{x - 2}\right)$

(e) $m(x) = \dfrac{\cos x}{x^2 + 1}$

Problem 3 *Classify the discontinuity at the given number as removable, jump or infinite. If it is removable, give the value that repairs it.*

(a) $f(x) = \dfrac{x^2 - 16}{x - 4}$ at $a = 4$

(b) $f(x) = \dfrac{|x - 5|}{x - 5}$ at $a = 5$

(c) $f(x) = \dfrac{x + 2}{x^2 - 4}$ at $a = 2$

(d) $f(x) = \dfrac{x + 2}{x^2 - 4}$ at $a = -2$

(e) $f(x) = \dfrac{\sqrt{x} - 3}{x - 9}$ at $a = 9$

Problem 4 *Piecewise functions: find the constant that makes f continuous everywhere.*

(a) $f(x) = \begin{cases} 3x + c, & x \le 1, \\ x^2 - 2, & x > 1 \end{cases}$

(b) $f(x) = \begin{cases} cx^2, & x < 2, \\ x + 4, & x \ge 2 \end{cases}$

(c) $f(x) = \begin{cases} \dfrac{x^2 - 1}{x - 1}, & x \ne 1, \\ c, & x = 1 \end{cases}$

(d) $f(x) = \begin{cases} x + a, & x < 0, \\ x^2 + b, & 0 \le x \le 2, \\ 5x - a, & x > 2 \end{cases}$ (find a and b)

(e) $f(x) = \begin{cases} \dfrac{\sin x}{x}, & x \ne 0, \\ c, & x = 0 \end{cases}$

Problem 5 *Limits by continuity. Evaluate, and name what makes each step legal.*

(a) $\lim\limits_{x \to 2} \dfrac{x^3 - 4x + 1}{x^2 + 1}$

(b) $\lim\limits_{x \to 3} \sqrt{x^2 + 7}$

(c) $\lim\limits_{x \to 1} \sin\!\left(\dfrac{\pi x}{2}\right)$

(d) $\lim\limits_{x \to 4} \sqrt{\dfrac{x^2 - 16}{x - 4}}$

(e) $\lim\limits_{x \to 2} \cos\!\left(\pi \, \dfrac{x^2 - 4}{x - 2}\right)$

Problem 6 *The Intermediate Value Theorem.*

In (a)–(d), show that the equation has a solution in the given interval. Write the argument in full: name the function, say why it is continuous, give the two values, and state the conclusion.

(a) $x^3 + 2x - 4 = 0$ on $(1, 2)$

(b) $x^4 + x - 3 = 0$ on $(1, 2)$

(c) $\cos x = x$ on $(0, 1)$

(d) $\sqrt{x + 1} = x^2$ on $(1, 2)$

(e) True or false: "$f(0) = -1$ and $f(2) = 3$, so $f(c) = 0$ for some c in $(0, 2)$." Justify, or give a counterexample.

Part II · Limits at infinity and horizontal asymptotes (Section 3.4) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. If $\lim_{x\to\infty} f(x) = 3$, then $y = 3$ is a horizontal asymptote of the graph of f.
2. The graph of a function can never cross its horizontal asymptote.
3. A function can have two different horizontal asymptotes.
4. The graph of a rational function can have two different horizontal asymptotes.
5. $\lim_{x\to\infty}(x^2 - x) = 0$, because $\infty - \infty = 0$.
6. $\displaystyle\lim_{x\to-\infty} \frac{1}{x^3} = 0$.
7. For $x < 0$, $\sqrt{x^2} = x$.
8. $\displaystyle\lim_{x\to\infty} \sin x = 0$.
9. If $\lim_{x\to\infty} f(x) = \infty$, then the graph of f has no horizontal asymptote as $x \to \infty$.
10. $\displaystyle\lim_{x\to\infty} \left(\sqrt{x^2+3x} - x\right) = 0$.

Problem 7 *Rational functions at infinity. Divide by the highest power of x in the denominator.*

(a) $\displaystyle\lim_{x\to\infty} \frac{4x^2 - 3x + 1}{2x^2 + 5}$

(b) $\displaystyle\lim_{x\to\infty} \frac{2x + 7}{3x^2 - x}$

(c) $\displaystyle\lim_{x\to\infty} \frac{x^3 + 1}{4x^2 + x}$

(d) $\displaystyle\lim_{x\to-\infty} \frac{5x^3 - 2x}{1 - 3x^3}$

(e) $\displaystyle\lim_{x\to-\infty} \frac{x^2 + 1}{x^3 - x}$

Problem 8 *Square roots at infinity: the sign trap and the conjugate. (Write $\sqrt{x^2} = |x|$ before you divide.)*

(a) $\displaystyle\lim_{x\to\infty} \frac{\sqrt{9x^2 + 2}}{4x - 1}$

(b) $\displaystyle\lim_{x\to-\infty} \frac{\sqrt{9x^2 + 2}}{4x - 1}$

(c) $\displaystyle\lim_{x\to\infty} \left(\sqrt{x^2 + 5x} - x\right)$

(d) $\displaystyle\lim_{x\to\infty} \left(\sqrt{x^2 + 3x} - \sqrt{x^2 - x}\right)$

(e) $\displaystyle\lim_{x\to-\infty} \left(x + \sqrt{x^2 + 4x}\right)$

Problem 9 *Find all asymptotes, horizontal and vertical. (A hole is not an asymptote.)*

(a) $y = \dfrac{3x + 1}{x - 2}$

(b) $y = \dfrac{x^2 - 1}{x^2 - 4}$

(c) $y = \dfrac{x - 3}{x^2 - 9}$

(d) $y = \dfrac{x}{\sqrt{x^2 + 1}}$

(e) $y = \dfrac{x^2 + x}{x - 1}$

Problem 10 *Infinite limits at infinity: what does the function do far out?*

(a) $\lim_{x\to\infty}\left(x^3-4x^2\right)$

(b) $\lim_{x\to-\infty}\left(x^3-4x^2\right)$

(c) $\lim_{x\to\infty}\left(x^2-x^4\right)$

(d) $\lim_{x\to-\infty}\left(2x^5+x^2\right)$

(e) $\lim_{x\to\infty}\left(\sqrt{x}-x\right)$

Problem 11 *Oscillation at infinity: when does the Squeeze Theorem help?*

(a) $\lim_{x\to\infty}\dfrac{\cos x}{x}$

(b) $\lim_{x\to\infty}\cos x$

(c) $\lim_{x\to\infty}\dfrac{x+\sin x}{x}$

(d) $\lim_{x\to\infty}\dfrac{\sin(x^2)}{\sqrt{x}}$

Problem 12 *The precise definition at infinity.* Recall: $\lim_{x\to\infty} f(x)=L$ *means: for every* $\varepsilon>0$ *there is a number* N *such that if* $x>N$ *then* $|f(x)-L|<\varepsilon$.

Prove each of the following from the definition. Scratch work first (find N), then the proof written forward.

(a) $\lim_{x\to\infty}\dfrac{1}{x}=0$

(b) $\lim_{x\to\infty}\dfrac{1}{\sqrt{x}}=0$

(c) $\lim_{x\to\infty}\dfrac{3x+1}{x}=3$

(d) $\lim_{x\to\infty}x^2=\infty$ (for every $M>0$ there is N with $x>N\Rightarrow x^2>M$)

Challenge *(for those who finish early, or at home)*

(a) Show that every polynomial of *odd* degree has at least one real zero. (Hint: what are the limits at $+\infty$ and at $-\infty$? Then apply the Intermediate Value Theorem on a large enough interval.)

(b) Let f be continuous on $[0,1]$ with $0\le f(x)\le 1$ for every x. Show that f has a *fixed point*: some c in $[0,1]$ with $f(c)=c$. (Hint: apply the IVT to $g(x)=f(x)-x$.)